

Chip-scale electrically driven superconducting coherent photon sources for quantum information processing

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Lasers and superconductors both rely on macroscopic quantum coherence. In lasers, coherence arises through stimulated emission, whereas in superconductors, it results from spontaneous symmetry breaking of the quantum ground state. This coherence underpins superconducting devices such as Josephson junctions, which generate electromagnetic radiation under an applied voltage. The emission frequency is governed by the superconducting energy gap, allowing operation across the microwave to terahertz regimes. High-temperature superconductors extend this range up to 15 THz. Furthermore, hybrid superconductor–semiconductor platforms, such as superconducting light-emitting diodes, open pathways to optical photon generation, including single- and entangled- photon emission. Here, I highlight how superconducting materials and Josephson junction-based hybrid devices enable compact, chip-scale, electrically driven, electrically tunable, power-efficient coherent light sources that span the microwave, millimetre-wave, terahertz and optical regimes and explore their potential for emerging quantum technologies and scalable quantum information processing.

The electrons in superconductors condense into Cooper pairs, resulting in entangled two-electron singlet states. The coherence and entanglement features of superconductors can be used to manipulate quantum states¹. For instance, quantum communication networks would benefit from compact and efficient solid-state photon sources capable of emitting single photons or entangled photon pairs on demand, thereby transforming high-speed, unconditionally secure information². At the heart of these quantum networks are quantum bits (qubits), the key components of information used for storage, processing and communication. Several qubit solutions have been suggested, but superconducting qubits operating at microwave frequencies are among the leading technologies due to their scalability and potential integration into large-scale semiconductor quantum circuits³. They enable coherent qubit operations in electrical circuits built using routine lithography. Developing superconducting circuits is therefore critical for creating complex, but robust, quantum systems capable of supporting future communication and computing networks.

For example, on-demand microwave photon sources that can generate quantum signals with high efficiency and spectral purity will facilitate the realization of distributed quantum computing⁴. High-speed and high-bandwidth terahertz (THz) links enable ultrafast wireless quantum communication between quantum computer systems⁵. Moreover, quantum cryptography is approaching real-world implementation, extending into sophisticated quantum information networks based on photon qubits⁶. This growth would require the reliable, efficient and effective creation, control and conversion of superconducting microwave qubits to flying photon qubits (and vice versa), which would allow continuous information exchange over today's optical fibre communication networks⁷. However, the underlying technology for the control and entangling of superconducting qubits and photon qubits is still being developed.

Quantum entanglement occurs when the state of a two-photon system cannot be expressed as a direct product of individual photon states. One key characteristic is that the states of the two photons are

correlated but remain undetermined until measurement⁸. When one photon's state is determined by measurement, the second photon's state is instantly fixed, regardless of the distance between them. This non-local correlation is a distinguishing characteristic of quantum entanglement. Concerning photon qubits, entangled photon pairs have been produced by employing parametric downconversion, resulting in a high degree of entanglement⁹. However, parametric downconversion is based on nonlinear materials driven by lasers, which poses some issues¹⁰. First, it results in Poisson-distributed photon pairs with unpredictable timings and quantities. This randomness makes it challenging to generate single photon pairs on demand, hindering the practical application of parametric downconversion in entangled photon sources. Second, the laser-driven entanglement method makes the scalability and integration of quantum circuits complicated. Concerning solid-state photon sources, electron–hole recombination in semiconductors leads to the emission of visible light¹¹. Exciton emissions have been used to generate single photons in electrically driven semiconductor quantum dots (QDs), with higher-level excitations yielding correlated biexciton emissions¹².

Despite major global efforts, demonstrating efficient quantum entanglement between biexciton and exciton photon pairs remains a challenge. The major barrier to quantum entanglement is the energy splitting of intermediate exciton states, which determines biexciton emissions¹³. This splitting arises from the in-plane anisotropy of QDs, which is caused by crystallographic surface anisotropy and electron–hole exchange interactions. Despite efforts to establish entangled photon pair creation via biexciton–exciton cascade recombination, the level of entanglement remains modest¹⁴. Photon pairs formed by this cascaded process have time separations on the order of the exciton lifetime, which is about 1 ns or less in QDs¹⁰. As a result, the quantum entanglement of these pairs is heavily dependent on the coherence times of exciton and biexciton emissions, which are often shorter than their lifetimes. Furthermore, these recombination processes encounter difficulties in obtaining identical photon pairs. Biexciton and exciton radiative recombinations have different lifetimes, making it difficult to achieve the ideal temporal overlap of their waveforms. The cascade emission mechanism introduces a time delay between biexciton and exciton emission, which may result in a spin flip of the intermediate exciton states, causing decoherence between the photon states. This creates a considerable challenge for photon qubit operations, as the capability to produce indistinguishable optical photons or photon pairs on demand is vital for quantum information applications¹².

The combination of superconducting Josephson junctions (JJs) and semiconductor QDs within a single hybrid nanostructure represents a possible and viable solution¹⁰. QDs allow the integration of photon-based technologies with solid-state systems such as electron spin qubits in semiconductor devices, or charge or flux qubits in superconducting circuits. The integration of semiconducting QDs with superconducting JJs enables the development of superconducting light-emitting diodes (SLEDs), which improve the semiconductor's light-emission capability via the intrinsic coherence of the superconducting state⁷. This hybrid technique leverages the strengths of both platforms to realize practical quantum information processing. For example, two-photon emission can be generated using Cooper-pair-related radiative recombination in a QD¹⁵. Here I discuss chip-scale superconducting coherent photon source technologies and their rich potential as an important integrated quantum hardware to advance quantum information processing and communication networks.

Superconductor microwave photon sources

Superconductor microwave light sources emit single microwave photons on demand and are essential for various applications including quantum metrology, communication, computation and information processing^{3,4}. They benefit from intrinsic superconductor

characteristics such as simple fabrication, integration, compactness, fast operation speeds, low dissipation, strong nonlinearity, low-frequency dispersion, high efficiency and long coherence. Their main components are (1) superconducting materials, such as niobium (Nb) and niobium nitride (NbN), which possess zero direct current (d.c.) and low radiofrequency (rf) electrical resistance below the superconductor critical temperature. (2) JJs, integral building blocks in superconducting circuits, which are the active component in generating coherent microwave radiation. A JJ consists of two superconducting electrodes separated by a thin non-superconducting (for example, insulating or semiconducting) barrier that operates as a voltage-to-frequency converter based on the alternating current (a.c.) Josephson effect¹⁶. (3) Microwave resonators, such as coplanar waveguide or planar transmission line resonators, are designed to resonate at certain microwave frequencies¹⁷, allowing the selective amplification or filtering of the desired signal generated by the JJs.

Superconductor microwave light sources maintain quantum coherence by preserving the phase relationship among microwave photons. Multiple techniques have been developed to generate stable and controllable coherent quantum microwave signals on the chip using superconducting devices and circuits^{4,18–31}. For instance, on-chip, superconducting single-photon sources can be realized in a circuit quantum electrodynamics design that incorporates a microwave transmission line cavity for the amplification of the spontaneous emission of a single qubit, as demonstrated in Fig. 1a (ref. 4). A superconducting qubit's spontaneous radiation generates a photon that behaves similarly to a flying qubit. This is because the state of a qubit in any superposition state can be translated to a superposition of zero and one photons. As a result, information can be transmitted from a stationary to a flying qubit by sending both quantum phase and amplitude across the chip during the emission. This single-microwave-photon generator operates using the Purcell phenomenon, in which a qubit is firmly coupled to a cavity. During the decay time of each control pulse-excited qubit, only one photon is emitted. The spontaneous emission rate of this microwave source can be identified from the Jaynes–Cummings Hamiltonian, $H = \hbar\omega_a\sigma_z/2 + \hbar\omega_c(a^\dagger a + 1/2) + \hbar g(a^\dagger\sigma + a\sigma_z)$, where the first two terms define a qubit with frequency ω_a described by Pauli operators σ_x , σ_y and σ_z and raising and lowering operators σ_+ and σ_- , and a single cavity mode of frequency ω_c defined by the photon creation and annihilation operators $a^\dagger$ and a ; $\hbar$ is the reduced Planck constant. The last term shows the coupling g between the qubit and photon, containing both the individual qubit and photon eigenstates.

If detuning between two frequencies in the system $\Delta = \omega_c - \omega_a \gg g$ (far detuned system), the qubit gains a minor photonic component of the wavefunction with a magnitude g/Δ , which reveals another cause of decay for the qubit. This is because the photonic component of the qubit can radiate at the cavity decay rate k , leading to a new qubit decay rate. Therefore, the qubit will be an efficient photon source when the new decay rate $\gamma_k = (g/\Delta)^2 k$ dominates over other non-radiative decay rates, $\gamma_k > \gamma_{\text{NR}}$. The lasing effect can also be realized with a JJ charge qubit, as a single artificial atom embedded in a superconducting resonator (the device shown in Fig. 1b), when strongly and controllably coupled to the resonator modes¹⁸. Chip-scale coherent microwave lasers are realized using the a.c. Josephson effect by employing a JJ that is strongly coupled with a multimode superconducting cavity^{19,20}. Applying a d.c. voltage across a JJ produces an a.c. supercurrent that oscillates at microwave frequencies or higher, depending on the superconducting energy gap. As a result, a d.c. voltage-biased JJ can be used to generate efficient coherent microwave photons, and the circuit's nonlinearity allows the active transition of higher-order Josephson frequencies to the cavity's fundamental mode.

A single JJ works similarly to a two-level atom. When the voltage difference across the JJ reaches a point where twice the voltage equals a multiple of the cavity frequency ($2eV_b = \hbar f$, where $2e$ denotes the charge on a Cooper pair, V_b is the bias voltage, $\hbar$ is Planck's constant

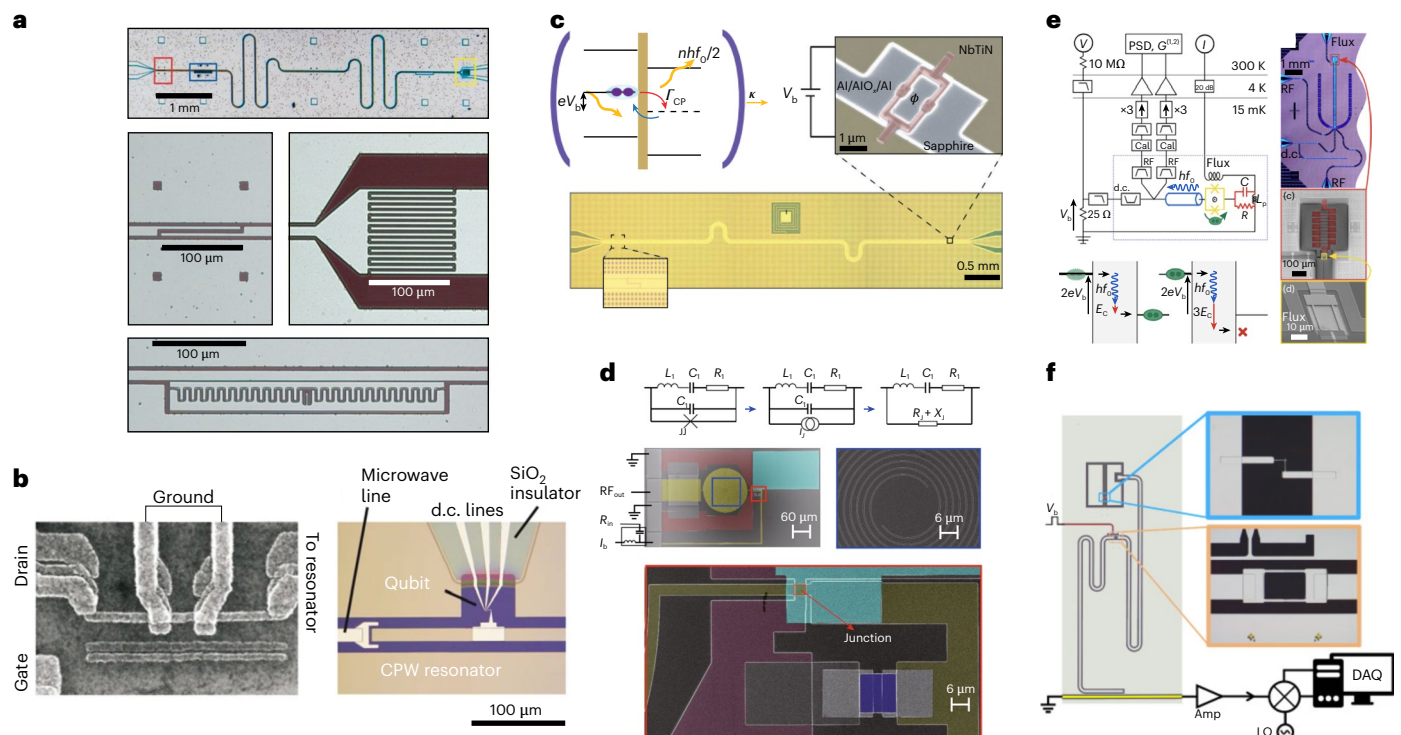


Fig. 1 | Superconductor microwave photon sources. **a**, Single microwave photon emitter that uses a superconducting circuit quantum electrodynamics architecture⁴. An Al/SiO₂/Si transmission line cavity lies between two capacitors (input, red; output, yellow). The transmon qubit (blue) is positioned at a voltage antinode. **b**, Components of an artificial-atom maser shown in a scanning electron microscopy (SEM) image of the qubit (left) and a micrograph of a Nb coplanar waveguide (CPW) resonator (right)¹⁸. **c**, Josephson-effect microwave laser. A circuit schematic, SEM image and optical image showing a d.c. SQUID and $\lambda/2$ CPW element are shown. Emission is voltage-driven (V_b)¹⁹. h , Planck's constant. **d**, Low-noise microwave source based on a Nb JJ coupled to a spiral

resonator. The ground plane is shown in pink, the bias line and resonator in yellow and the shunt capacitor in cyan²⁰. **e**, Antibunched photon source. An optical image and SEM images of a SQUID with vertical NbN/MgO/NbN JJs and an RC circuit (shown in the red box) exhibiting inelastic Cooper-pair tunnelling²¹ are shown on the right. **f**, A deterministic quantum source with a transmon qubit and SQUID. Optical micrographs (right) and the homodyne detection set-up²² (bottom) are also shown. Figure reproduced with permission from: **a**, ref. 4, Springer Nature Limited; **b**, ref. 18, Springer Nature Limited; **c**, ref. 19, AAAS; **d**, ref. 20, Springer Nature Limited; **e**, ref. 21 under a Creative Commons license CC BY 4.0; **f**, ref. 22 under a Creative Commons license CC BY 4.0.

and f is the frequency), an imperative phenomenon arises: the system's two energy levels for the Cooper pairs become clearly defined. This interaction can result in either spontaneous or stimulated emission, and the absorption of photons, within the cavity^{4,18,19}. The d.c. voltage bias causes the transfer rate of Cooper pairs across the JJ (I_{CP}) to exceed the cavity lifetime (κ) of microwave photons, resulting in lasing. In this situation, microwave photons are frequently absorbed and reemitted by the JJ, a process similar to stimulated emission in optical or atomic laser systems. The reabsorption and re-emission sequence boosts photon coherence, comparable to light amplification through stimulated emission in lasers by interacting with excited atoms. These photon-JJ interactions contribute to coherent microwave radiation. For instance, coupling a d.c. superconducting quantum interference device (SQUID) with a $\lambda/2$ superconducting microwave coplanar waveguide resonator (where λ is the wavelength) results in characteristics including injection locking and frequency-comb generation at microwave frequencies¹⁹. This circuit, which uses an on-chip inductor at the cavity's electric field node, as well as two coupling capacitors at the input and output of the cavity, can produce injection-locked linewidths as small as ≤ 1 Hz, comparable to the output of advanced optical lasers. It can be configured to operate in either linear or nonlinear regimes. In the linear regime, the JJ and cavity are weakly coupled, which leads to spontaneous microwave emission. The nonlinear regime enables strong coupling between the JJ and the cavity, resulting in stimulated microwave emission. The tunability between regimes offers a versatile platform for manipulating photon emission cycles, as switching from weak to strong coupling allows several forms of operation within the

microwave domain, including quantum amplification, lasing and even entanglement creation.

A coherent continuous-wave microwave source can also be built on-chip by combining a JJ with a superconducting spiral resonator as shown in Fig. 1d (ref. 20). The properties of this architecture follow the theoretical paradigm of a perturbative capacitively shunted JJ, which works like the gain element. This design provides control over the microwave output, using the nonlinear dynamics of the JJ to produce stable, coherent continuous-wave microwave radiation. Expanding on these platforms, coherent microwave sources capable of emitting on-demand antibunched microwave radiation can be developed through coherent control of Josephson qubits with regulated microwave pulses, as depicted in Fig. 1e (ref. 21). This approach enables the generation of non-classical microwave radiation. A simpler set-up uses a d.c. voltage-biased JJ, which can work as a bright source of quantum microwave radiation. These sources feature antibunching behaviour in the single-photon range, generating individual photons into a microwave resonator using inelastic Cooper-pair tunnelling. This process is controlled by a high-impedance resistor-capacitor (RC) circuit, which ensures that photon emission is coherent.

Furthermore, superconducting quantum circuits can be used to create deterministic microwave quantum light sources that emit propagating single photons, as well as time-bin encoded photonic qubits and qudits. The frequency of emitted photons in these on-chip sources (Fig. 1f) can be tuned in situ by up to 200 MHz. This tunability enables fine control while preserving the integrity of the propagating photonic qubits, particularly when using a time-bin encoding technique²².

Superconductor coherent millimetre-wave and THz sources

With superconductors that use the Josephson effect, it is also possible to generate coherent millimetre waves and THz photons on a chip, typically between 100 GHz and 10 THz (refs. 32,33). JJs naturally convert d.c. voltages to high-frequency electric currents, resulting in electromagnetic radiation that crosses the THz gap, where 1 mV equals about 0.483 THz. While the emission power from a single junction is normally in the picowatt regime, a device that contains multiple junctions emitting in phase at the same frequency can dramatically increase the output power, reaching microwatt levels³⁴. The frequency of generated THz radiation is determined only by the material's superconducting energy gap. In superconductors with low transition temperatures (T_c), the frequency is often limited to the millimetre-wave (sub-THz) range. However, in high-transition temperature (high- T_c) superconductors, the frequency can reach around $f = 15$ THz. Because of their broad frequency range, these high- T_c superconductors are especially attractive for applications that require higher-frequency THz radiation³⁵.

Interconnected JJs may self-synchronize with a collective radiation field and emit coherently via a quantum-stimulated emission mechanism involving Cooper pairs and photons. The underlying physical mechanism of synchronization is identical to the quantum description of a group of superradiant atoms in a resonating cavity³⁶. The output power scales with the square of the number of active junctions N . In general, such an N^2 dependence indicates coherence in a system of N oscillators. The connection between a single JJ and a quantum radiating two-level atom demonstrates that spontaneous and stimulated emission and absorption can occur in JJs. Two-dimensional (2D) JJ arrays made of underdamped Nb–Al–AlO_x–Nb junctions, which are macroscopic and non-identical objects with no shunt resistors, coupled to a high-quality factor (Q) cell resonance (Fig. 2a) emit quantum coherent radiation at $f = 150$ GHz in the same way that atoms in a laser emit. The emission frequency is equivalent to a high- Q resonance in the structure formed by the array and the ground plane. This resonance stimulates sharp, self-induced steps in the current–voltage characteristics of the array. The power coupled to the detector is measured as a function of the number of junctions that are biased at the resonant step. In such a scenario, no detectable power is observed if the number of active junctions is below a critical threshold (N_T). This behaviour is analogous to the threshold phenomenon in lasers (where coherent radiation appears only when the gain exceeds the losses, indicating the presence of population inversion), underlining their potential as efficient coherent millimetre-wave and THz radiation sources³⁶. Above N_T , the detected power scales with the square of the number of active junctions, reaching a d.c.–a.c. power conversion efficiency of nearly 17%. In comparison, arrays that synchronize via classical coupling processes have a maximum a.c.–d.c. conversion efficiency of only 1%. The emission power from a single JJ is relatively small, but it can be largely enhanced in a superradiant manner by using either planar or vertical JJ

arrays with several synchronized junctions ($N \gg 1$). Synchronization is often facilitated by a resonant electromagnetic mode, such as a cavity mode (a standing electromagnetic wave), which can be established either within an external cavity or within the dielectric substrate^{37,38}. To achieve an emission power at the milliwatt level, very large arrays (such as that shown in Fig. 2b) containing approximately $N \approx 10^4$ – 10^5 JJs would be required^{39,40}. Considering the achievable integration density, such arrays would typically have a size of $L \approx 1$ cm, which is significantly larger than λ even at millimetre-wave frequencies⁴¹. However, as the size of the planar 2D array increases, so does the complexity of achieving synchronization. The challenge originates from variations in junction parameters, which statistically increase with N , as well as unequal environmental conditions for the inner and outer JJs. In addition, large phase delays along the array hinder the synchronization process. In the non-resonant scenario, synchronization can be accomplished using a unidirectional travelling wave along the array, which functions in a way similar to a travelling wave (Beverage) antenna. The primary feature of this travelling-wave domain is a strong forward–backwards asymmetry in the radiation pattern, with the majority of the power radiated in the direction of the wave's propagation. This behaviour differs significantly from the resonant standing wave scenario, in which the radiation pattern is symmetric by definition. As a result, the form of the radiation pattern makes it easy to distinguish between these two operational modes. The previously described planar 2D JJ arrays were artificially produced structures, the large-scale fabrication of which is quite challenging. In comparison, 1D arrays of JJs naturally grow in layered high- T_c superconductor Bi₂Sr₂CaCu₂O_{8.6} (BSCCO). Devices based on BSCCO are emerging compact sources of coherent continuous-wave electromagnetic radiation in the millimetre-wave and THz frequency bands. In BSCCO, the junctions are naturally identical^{34,35}. They are evenly spaced with two junctions per unit-cell c -axis edge length of 1.533 nm, forming stacks of JJs (Fig. 2c). The superconducting CuO₂ layers of BSCCO, coupled with the intrinsic Josephson effect, can maintain high voltages across junctions. Owing to the large superconducting energy gap, BSCCO devices can produce powerful, coherent radiation spanning the THz gap with emission powers of free-space radiation of several hundred microwatts and emission linewidths as low as 6 MHz at 600 GHz (refs. 32,42). The efficient excitation of high-frequency currents is an important consideration in the development of superconducting sources. This can be accomplished by triggering resonances such as the Fiske resonance using a junction bias. Similar to a laser cavity, the stimulation of an electromagnetic cavity resonance within the sample generates a macroscopic coherent state, causing a large number of junctions to oscillate in phase^{42–44}. Modern flux-flow oscillators provide microwatt-level powers at frequencies ranging from 400 GHz to 600 GHz, with emission linewidths as small as ~0.6 MHz. 1D or 2D arrays of phase-locked JJs result in much larger emission powers; for example, an array of 1,968 junctions delivers 160 μ W at a frequency of 240 GHz. Phase locking is critical, and it can

Fig. 2 | Superconductor coherent millimetre-wave and THz photon sources.

a, A 3×36 array of Nb–Al–AlO_x–Nb underdamped Josephson tunnel junctions designed for millimetre-wave generation. The optical image below shows the ~470 μ m array length³⁶. **b**, Top view of a fabricated linear planar JJ array and a meander structure containing over 1,500 JJs used to develop superconducting millimetre-wave sources⁴¹. **c**, Schematic of intrinsic JJs in BSCCO crystals, composed of alternating superconducting and insulating layers. Optical images of grown BSCCO crystals prepared for THz emission^{35,55} are shown below. **d**, BSCCO THz sources with engineered cavity geometries that enable various fundamental modes for narrowband and broadband THz output⁴³. **e**, Optical image of a BSCCO whisker THz source array with 12% radiation efficiency at 3 K⁷⁰. **f**, Image (left) and schematic (right) of a THz BSCCO device array exhibiting spontaneous frequency shifts and phase delays via Josephson plasmons⁷⁴. **g**, BSCCO device mounted on a sapphire lens, emitting tunable THz radiation from 0.15–1 THz (ref. 76). **h**, Micrograph of a BSCCO device generating 130 μ W

of coherent THz power at 0.456 THz at a 77.4 K bath temperature⁷⁷. **i**, On-chip BSCCO THz source with a gigahertz-scale frequency modulation bandwidth that can modulate 840–890 GHz carrier waves via d.c. and radiofrequency (RF) signals^{79,80}. **j**, Mutually synchronized BSCCO stacks with thermally conductive gold layers enabling scalable, power-efficient THz sources⁷⁸. **k**, Top left: performance comparison of solid-state on-chip THz emitters, showing BSCCO-based sources spanning the THz gap. Bottom left: imaging results for a dandelion leaf demonstrate application potential where a THz imaging set-up uses a BSCCO THz Josephson plasma emitter as a source of coherent THz radiation (right)⁸⁷. **JPE**, Josephson plasma emission; **Trans**, transmission. Figure adapted with permission from: **a**, ref. 36, APS; **b**, ref. 41, APS; **c** (top), ref. 35, AAAS; **c** (bottom), ref. 55, Springer Nature Limited; **d**, ref. 43, IOP; **e**, ref. 70 under a Creative Commons license CC BY 4.0; **f**, ref. 74, APS; **g**, ref. 76 under a Creative Commons license CC BY 4.0; **h**, ref. 77, APS; **i**, ref. 80, Springer Nature Limited; **j**, ref. 78, APS. Panel **k** courtesy of the authors of ref. 87.



Table 1 | Types of BSCCO THz device with different cavity shapes, dimensions and resonance modes

Superconducting THz cavity shape	Size (μm)	Height (μm)	Resonance mode
Rectangle	$45 \leq w \leq 100$, $120 \leq L \leq 400$	$0.8 \leq t \leq 2.0$	$\frac{c_0}{2n} \sqrt{\left(\frac{m}{L}\right)^2 + \left(\frac{p}{w}\right)^2}$
Square	$66.7 \leq L \leq 146$	$1.0 \leq t \leq 1.5$	$\frac{c_0}{2n} \sqrt{\left(\frac{m}{L}\right)^2}$
Cylinder	$33.9 \leq a \leq 61.5$	$1.0 \leq t \leq 1.2$	$\frac{c_0}{2\pi a n} \chi_{mp}$
Equilateral triangle	$100 \leq b \leq 260$	$1.0 \leq t \leq 1.5$	$\frac{2c_0}{3bn} \sqrt{m^2 + mn + n^2}$
Isosceles triangle	$75 \leq b \leq 120$	$1.0 \leq t \leq 1.5$	$\frac{c_0 \pi}{\sqrt{2}bn} m$
Regular pentagon	$58 \leq a \leq 60$	$1.0 \leq t \leq 1.2$	$\frac{c \chi_{mp}}{2\pi a n \pi_r}$

be accomplished by placing the junctions in a microwave stripline at half-wavelength intervals on a common ground plane linked to a resonant cavity or loading the array with an external inductor–capacitor–resistor circuit (LCR) resonance circuit. The radiation power of an array depends on the array geometry and the external circuitry. The power of a 1D series-connected array scales as N^2 (where N is the number of phase-locked junctions) for an unmatched load (for which the effective array resistance is smaller than the load resistance) and N for a matched load. The emission linewidth decreases as $1/N$ for all array types. Examinations of devices with various cavity geometries (from square^{45–47} to rectangular^{48–53}, cylindrical⁵⁴, triangular^{55–59} and pentagonal^{60,61}, as shown in Fig. 2d) fabricated using different techniques and under varying conditions reveal that the a.c. Josephson effect is the principal driving mechanism for the radiation in BSCCO THz sources. While cavity resonance (Table 1) contributes to increased radiation strength, the two mechanisms act together during the emission process^{37,38,62–64}. The combination of the a.c. Josephson effect and cavity resonance is critical to produce stronger and coherent THz radiation^{38,65–69}. In such THz sources the radiation power efficiency (RPE; the ratio of emitted to dissipated power) drops quickly, especially in the low THz region⁷⁰. The measured RPE for BSCCO single-crystal-based devices with both large and small mesas is below 1%, lower than the theoretical limit of 50%. An impedance mismatch between the device and the open space mostly causes this lower efficiency. In general, a low RPE results from poor impedance matching, as JJs are significantly smaller than wavelengths, with interlayer distances as small as 1 nm. As a result, JJs act as microdipoles with low far-field emissions. In the end, the radiation is supported by the device's passive (but larger) elements, such as the BSCCO crystal and electrodes, which operate as matching antennas. To ensure impedance matching, the emission must be optimized via precise microwave design. Self-heating is an important issue encountered in all cryogenic devices^{70,71}.

Given the limited cooling capacity of tiny cryostats (typically sub-watt at low temperatures), devices with a 1% RPE would be unable to output much more than 1 mW. The THz emission power (P_{THz}) of cryogenic sources is restricted by the cooling power (P_{cooling}) and RPE; $P_{\text{THz}} < \text{RPE} \times P_{\text{cooling}}$ must be satisfied. In the case of compact cryocoolers that have low cooling power, the best way to boost the emission power is to improve the RPE. THz sources based on BSCCO whisker-type crystals with intermediate-sized mesa structures, as shown in Fig. 2e, could provide a solution to these issues⁷⁰. Various approaches can be used to detect THz radiation, such as in situ detection with a device on the same whisker, on-chip detection with electrically isolated devices and off-chip detection with a bolometer. Furthermore, the radiative cooling process can be used to determine the absolute value of the emitted power. Whisker-based BSCCO THz sources can achieve RPEs

of up to 12%, resulting in an emission power in the far field of around $P_{\text{THz}} \approx 0.11$ mW. The increase in efficiency (an order of magnitude greater than equivalent devices produced on traditional BSCCO single crystals) is related to better impedance matching with free space. This can be attributed mainly to the turnstile antenna-like design of the devices, which reduces parasitic capacitance between the crystal and the electrode. By minimizing unnecessary capacitance, the device improves impedance matching with open space, allowing more effective THz radiation emission. One of the most efficient techniques for improving the radiation intensity is to synchronize several mesas to release radiation simultaneously^{32–34}. Once two mesas operate in series, the combined emission intensity surpasses the total of their separate outputs. For instance, coherent THz emission can be achieved in devices with up to three mesas coupled in parallel, with the intensity scaling roughly proportionally to the number of synchronized mesas squared^{72,73}. Experimental results show an emission intensity of 0.61 mW, demonstrating the efficiency of this method⁷³. The polarization of the radiation emitted from individually biased mesas, together with the coupling matrix that describes the interaction of Josephson plasma waves between mesas, provides insight into how each mesa contributes to the overall synchronized emission when multiple mesas are simultaneously biased. Quantitative measurements of the polarization of the output electromagnetic wave allow an estimation of how each mesa affects the overall synchronized radiation. Comparing the emission properties of parallel and series couplings of identical mesas (like the devices shown in Fig. 2f) is especially relevant, because the coupling matrix might evolve when identical bases are employed, potentially resulting in different synchronization dynamics^{74,75}. In an emission mesa, where a thousand stacked intrinsic JJs (IJJs) oscillate coherently as a macroscopic JJ, the common voltage bias caused by parallel couplings should create synchronized oscillations across the macroscopic junctions. On the other hand, the common current bias caused by series couplings provides useful information on the performance of a smaller number of stacked IJJs. Inter-mesa coupling becomes critical, especially when two or three mesas are used in common voltage bias or common current bias modes. Studying the polarization of the electromagnetic waves generated by the mesas, both in the case of single mesa emission and in coupled emission modes, provides important information about their synchronization dynamics, revealing how the individual oscillators phase-lock and contribute to coherent collective radiation. In the common voltage bias mode, the emission has a unimodal frequency spectrum, indicating strong coupling between the mesas. In contrast, the common current bias mode generates uncoupled emissions with a bimodal frequency spectrum. The coupled emission state is effectively a superposition of independent emissions of radiation that are influenced by inter-mesa couplings. The coefficients of this linear combination form the interaction matrices between mesas. These interaction matrices indicate the harmonically oscillating plasmons within the superconducting substrate that facilitate the interaction of the nonlinear Josephson plasma oscillations in the mesas. The principles underlying powerful THz emissions from different devices have been investigated, demonstrating that BSCCO-based devices can produce THz radiation with frequencies ranging from 1 to 11 THz (ref. 71), but the self-heating issue must be solved for these BSCCO emitters to generate THz waves at frequencies lower than 200 GHz. Self-heating causes hot areas in the stack, which reduce the voltage across the JJs. One approach to address this issue is using sandwich-structure devices that can boost the cooling efficiency⁷⁶. These developments have also increased the emission frequency beyond 2 THz. At frequencies below 200 GHz, Josephson plasma frequency (f_{pi}) and resonant cavity modes dominate the THz emission frequency. The f_{pi} is dependent on temperature and follows the relation $f_{\text{pi}}(T) \propto (j_c(T))^{1/2}$ with a cutoff emission frequency around $4f_{\text{pi}}$. Here, j_c refers to the JJ critical current density. At relatively cold bath temperatures (T_b), f_{pi} is saturated, resulting in a constant cutoff

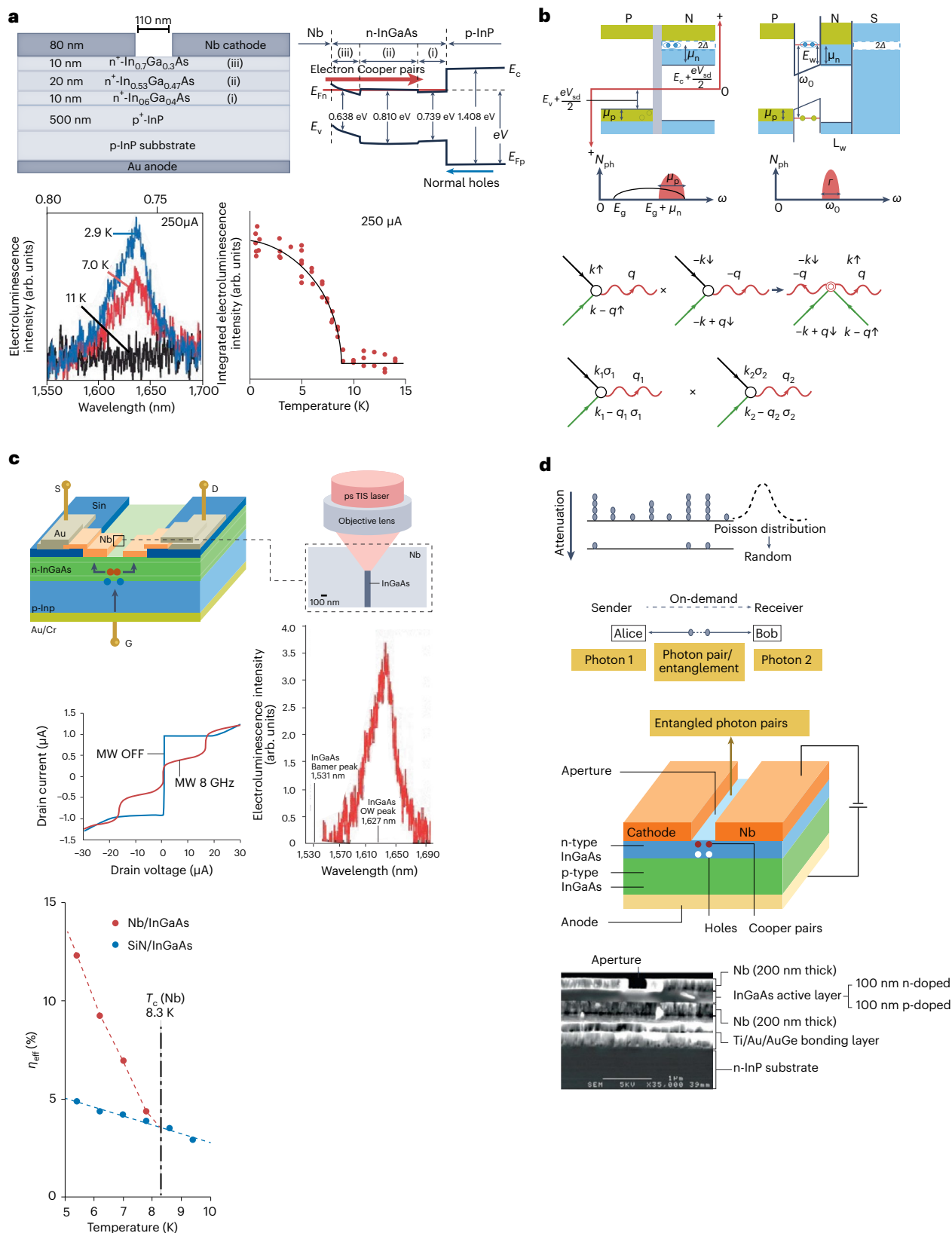


Fig. 3 | Superconducting light-emitting diodes. a, Schematic (left) and band structure (right) of a Nb/n-InGaAs/Nb SLED under forward bias that shows temperature-dependent electroluminescence enhanced below the superconducting transition temperature of Nb, which is evidence of Cooper-pair-induced superradiance⁹⁹. **b**, Theoretical energy diagram (top) and simulated emission spectra (middle) from forward-bias p-n junctions, modelling photon emission under superconducting conditions⁷. **c**, Top: diagram (right) and schematic (left) of a Cooper-pair SLED with a 110 nm Nb

electrode gap. Bottom: transport, electroluminescence and time-resolved photoluminescence measurements show Shapiro steps under microwave drive, and superconductivity-enhanced emission of the SLED⁹⁹. **d**, Schematic of photon generation mechanism (top) and SEM image (bottom) of selectively etched Nb apertures for deterministic entangled photon generation in SLEDs. Fabrication preserves the integrity of the InGaAs active layer¹⁰². Figure reproduced with permission from: **a**, ref. 8, Japan Society of Applied Physics/IOP; **b**, ref. 7, APS; **c**, ref. 99 under a Creative Commons license CC BY 4.0; **d**, ref. 102, IOP.

emission frequency at the bottom end of the spectrum. When T_b rises, f_{pl} falls, finally reaching zero when T_b reaches the critical temperature T_c , resulting in a lower low-frequency cutoff. Moreover, the activation of collective cavity resonances across the entire stack boosts the THz emission, with the emission frequency determined by a particular resonance mode. As a result, BSCCO stacks with low-frequency resonance modes can emit THz waves with adequate power in the lower frequency range. THz emitters with square gold–BSCCO–gold mesas (with $T_c = 88.6$ K) on sapphire substrates (Fig. 2g) can extend the emission frequency to $f = 0.15$ THz, generating a power output of 0.18 μ W at $T_b = 75$ K and 85 K (ref. 76). The operating temperatures of these devices notably exceed the boiling point of liquid nitrogen (77 K). In pushing the operation temperature of BSCCO devices higher than 77 K, an emitter with mesa of BSCCO doped with 0.16 holes per Cu atom with $T_c = 86.5$ K (Fig. 2h) generates THz radiation of 130 μ W coherent power at $f = 0.456$ THz at $T_b = 77.4$ K (ref. 77). This progress shows that implementing a BSCCO crystal with a different Bi/Sr stoichiometry could reach equivalent THz power levels at higher temperatures, resulting in a T_c higher than 86.5 K. This change may provide more thermal flexibility, making these devices better suited for emerging technologies and quantum applications that require stable operation at high temperatures⁵. Another approach to address the thermal conductivity issue is to use standalone stacks that are connected by gold layers and share a common gold layer⁷⁸. This design shown in Fig. 2j benefits from the thermal conductivity of gold being significantly higher than that of the BSCCO base crystals, while its patterning remains relatively straightforward. In this device, the two oscillators exhibit phase correlations within ± 0.4 GHz of their centre frequencies, ranging from 745 to 765 GHz. However, strong phase gradients in the beams radiated from both mutually locked and unlocked stacks seem to play a crucial role, probably reducing the detected emission power due to destructive interference. This effect may stem from higher-order cavity modes excited within individual stacks. The mutual interaction facilitated by the common gold layer could offer new possibilities for mitigating Joule heating, potentially enabling the synchronization of more stacks. This approach may also allow the synchronization of smaller stacks, which are less susceptible to strong phase gradients. BSCCO devices can generate coherent THz frequency signals with high-speed modulation reaching up to 40 GHz when 3 GHz sinusoidal waves on a d.c. bias of about 2 V are superimposed on 840 – 890 GHz carrier waves radiated by the device^{79,80}. As previously discussed, the JJs embedded within BSCCO emit THz waves with frequencies precisely proportional to V_b , following the relation $f = 2eV_b/h$. By modifying V_b , the Josephson plasma emission frequency can be modulated, allowing dynamic control over the output THz frequency, as schematically shown in Fig. 2i. The modulated signals generate discrete comb-like features, as defined by a Bessel function sum, with the spacing between each comb tooth corresponding to the modulation frequency⁷⁹. On-chip gigahertz-bandwidth frequency-modulated Josephson plasma emission requires precise control of the temporal behaviour of almost a thousand identical IJJs, which are naturally stacked in a vertical 1D array. As mentioned above, each IJJ is around 1.533 nm thick and has two junctions per unit cell that are evenly spaced^{32,33}. In BSCCO, superconductivity is limited to 2D planes held together by weak van der Waals interactions between the layers⁸¹. One of the most difficult challenges in implementing frequency modulation in BSCCO-based sources has been regulating the temperature changes produced by the higher modulation amplitude of the bias voltage. With precise electrical biasing and gigahertz synchronization of the phase emission of numerous IJJs of the device, the right choice of cavity geometry may address this issue. The detailed fundamental physics of electromagnetic wave radiation from JJs is explained in previous reports^{48,65}. For application purposes (Fig. 2k), BSCCO THz sources have several distinct benefits. These include chip integrability, very small footprints, low power dissipation, monolithic generation of linearly and circularly polarized coherent THz

waves, narrow emission linewidths of less than 10 MHz and the ability to cover a wide bandwidth across the THz gap^{82–87}. They also have a high RPE^{32,33,70}, which is promising for application domains from metrology^{80,81}, tomography⁸⁴ and imaging^{85–87} to future ultrafast communication⁵ and computing networks^{79,80}.

Superconductor single-photon and entangled-photon sources

Hybrid superconductor–semiconductor JJs can be engineered by integrating 1D or 2D semiconductors (for example InAs, GaAs and InGaAs) with superconducting materials (such as Al, NbN and Nb)^{88–91}. Superconductor-coupled p–n junctions and QDs in an LED configuration can realize solid-state entangled photon pair sources. Such SLED photon sources generate entangled single-photon pairs on demand, one at a time, through the injection of either electron or hole Cooper pairs into a semiconductor QD. The strong correlations between electron–electron or hole–hole pairs within the corresponding Cooper pairs will result in enhanced simultaneous recombination into photon pairs within the QD. The SLED source theoretically proposed by Hanamura⁹² is based on the superradiance effect arising from electron and hole Cooper-pair recombinations, in which Cooper pairs recombine simultaneously without going through intermediate exciton states in semiconductor QDs¹⁰. This concept was experimentally realized by Hayashi and colleagues⁹³, who demonstrated Cooper-pair injection from the superconducting Nb electrodes into semiconductors forming Nb/n-InGaAs/Nb vertical JJ through the proximity effect, showing that electron Cooper pairs could tunnel through 120 -nm-thick InGaAs layers on a p-type InP (p-InP) substrate, as shown in Fig. 3a. When electron Cooper pairs are forward biased, they enter the active layer of InGaAs and interact with normal holes in p-InP, resulting in a considerable electroluminescence (EL) increase below the superconducting transition temperature^{8,94}.

One proposed SLED source is based on the recombination of electron Cooper pairs and normal holes; the luminescence of a Cooper pair is theoretically investigated in detail in ref. 7, and the schematic of the device is shown in Fig. 3b. Recombination between electron and hole Cooper pairs is more challenging to realize experimentally in direct-bandgap semiconductors⁹⁵. Another type of Josephson LED or SLED was later theoretically proposed, in which a voltage-biased superconducting p–n junction that contains a semiconductor QD emits photons in the optical range⁹⁶. The suitability of the proposed Josephson LED for the on-demand production of entangled photon pairs and two-qubit gate quantum manipulation was theoretically explored⁹⁷. A proposal for the superconducting device is predicted to generate visible laser light at half the Josephson generation frequency, with the optical phase synchronized with the superconducting phase difference. This device contains two single-level QDs implanted in a p–n semiconducting heterostructure and enclosed by a cavity that enables a resonant optical mode as depicted in Fig. 4a (ref. 98). Contrary to conventional LEDs, where electrons and holes are injected incoherently through drift or diffusion, resulting in incoherent radiative recombination below the lasing threshold, SLEDs inject electrons as Cooper pairs in a condensed coherent state, significantly transforming the emission process. Furthermore, in Nb/n-InGaAs SLEDs, the current–voltage characteristics show both d.c. and a.c. JJ behaviours. In such a device, the radiative recombination lifetime τ_{rad} significantly reduces, but the internal quantum efficiency η_{eff} (which is determined by τ_{rad} and the non-radiative recombination lifetime (τ_{nonrad})) and follows the equation $\eta_{eff} = \tau_{rad}/(\tau_{total})$ increases below the superconducting critical temperature, as shown in Fig. 3c. The recombination lifetime can be calculated by $\tau_{total} = (\tau_{rad}^{-1} + \tau_{nonrad}^{-1})^{-1}$ (refs. 99–102). The gate voltage applied to the normal electrode on the p-type semiconductor side alters the transport characteristics between two superconducting electrodes on the n-type semiconductor side of the LED. This modulation shows good sensitivity to the radiative recombination process¹⁰⁰. The penetration of Cooper pairs into the semiconductor depends on the size, depth

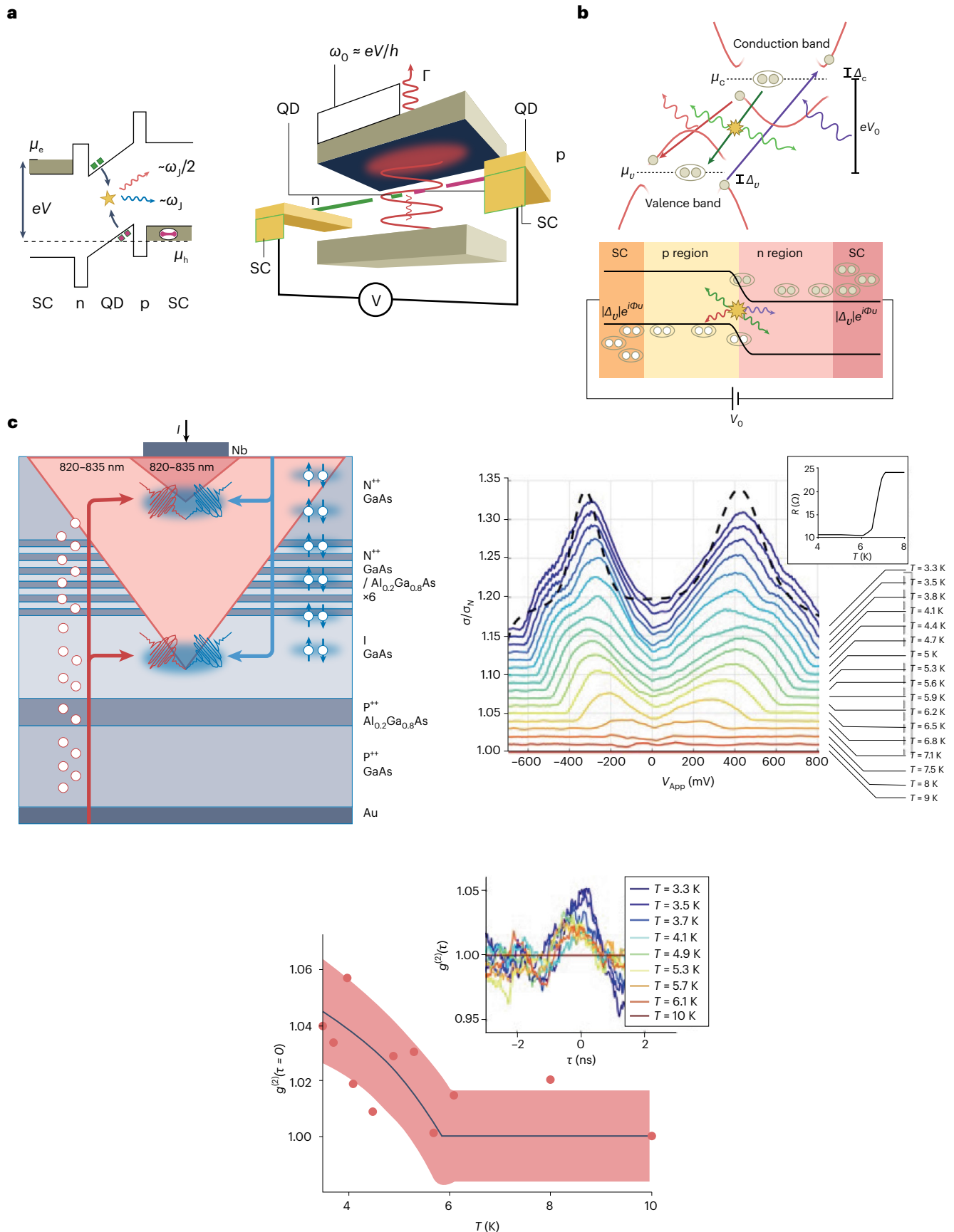


Fig. 4 | Superconducting light-emitting diodes. a, Theoretical model of a Josephson LED emitting light at half the Josephson frequency based on coherent pair tunnelling⁹⁸. **b**, Concept of a forward-bias SLED with p–n junction coupled to superconducting leads, enabling non-classical light emission¹¹⁰. **c**, Top left: schematic and emission zones of a Nb/GaAs/AlGaAs superlattice SLED, showing

spatially distinct electroluminescence origins. Photon coincidence ($g^{(2)}(\tau)$) and Andreev reflection measurements support superconductivity-driven photon pair correlations¹⁵. Figure reproduced with permission from: **a**, ref. 98, APS; **b**, ref. 110, APS; **c**, ref. 15 under a Creative Commons license CC BY 4.0.

and thickness of the superconducting cathode, affecting the emission efficiency (as demonstrated in Fig. 3d). Wider apertures increase the photon extraction efficiency, while narrower apertures improve Cooper-pair injection. Thinner superconducting layers result in lower critical temperatures and critical currents I_c , indicating that a thicker superconductor layer strengthens the proximity effect and improves the device's overall performance¹⁰². The electron Cooper pairs, injected into the semiconductor via the proximity effect, can actively contribute to interband transitions, accelerating the photon generation process¹⁰³. The fundamental of Cooper-pair injection in SLEDs is such that an electron with momentum (k) and spin ($\uparrow$) creates a spin-singlet state with another electron with momentum (k) and spin ($\downarrow$). This pair recombines with a pair of holes, one with momentum $k - q$ and spin ($\downarrow\downarrow$), and another with momentum $k + q$ and spin ($\uparrow\uparrow$). This process creates two indistinguishable photons, each with momentum q . The spin-singlet electrons represent an electron Cooper pair, and the entire process illustrates their recombination with the two holes (as illustrated in Fig. 3b). The radiative recombination rate follows the equation $W = 1/\tau_{\text{rad,super}} = A(\Delta^2(T)/T)\exp[(-2L_w)\xi_N(T)]$, where $\tau_{\text{rad,super}}$ is the minority carrier (hole) lifetime due to their radiative recombination with electron Cooper pairs in the active layer, $\Delta(T)$ is the temperature-dependent half of the superconducting energy gap, the term $\exp[(-2L_w)\xi_N(T)]$ accounts for the proximity effect with distance L_w from the superconductor–semiconductor interface to the active layer and ξ_N is the coherence length of Cooper pairs in the semiconductor. The SLED transient lifetime is $I_{\text{EL}}(t) = J_0(\eta_{\text{int}}\eta_{\text{det}}\exp(-t/\tau_{\text{CR}}))$, where J_0 is the carrier injection rate, τ_{CR} is the circuit CR constant and η_{int} and η_{det} are internal quantum efficiency and the external photon detection efficiency of the SLED, respectively. Transient photon emission from SLEDs depends on multiple parameters, such as the diode bias, the built-in electric field across the p–n junction, the width of the depletion layer and the injection of electron Cooper pairs and holes. The total number of photons emitted from SLEDs during the transient decay can be calculated as $N_{\text{photon}} = \int_0^\infty I_{\text{EL}}(t)dt = I_{\text{EL}}(t=0)\tau_{\text{LED}}$, where I_{EL} is the electroluminescence intensity, and τ_{LED} is the LED lifetime (refs. 104–107). A p–n junction between two superconductors can be used to emit squeezed light, form entangled photon pairs and influence photon density correlations. This squeezing event happens between modes with opposite momenta and is caused by the transmission of electronic coherence from Cooper pairs to emitted photons. The phase difference between the two superconductors allows modulation of the squeezing angle. This squeezed LED arrangement uses the macroscopic coherence of the superconductor to alter photon coherence during a two-photon pumping process¹⁰⁸. It was predicted that the recombination of Cooper pairs in semiconductors could be used as a natural source of polarization-entangled photons, leveraging the inherent angular momentum entanglement in the superconducting state without the need for isolated emitters such as single atoms or QDs¹⁰⁹. SLEDs can also potentially produce two-mode squeezed light, as shown in Fig. 4b, with the squeezing angle controlled by the superconducting phase difference between the conduction and valence bands of a forward-bias semiconducting p–n junction¹¹⁰. In addition to the proposed SLED devices discussed earlier that feature Josephson effects and enhanced electroluminescent light below T_c , the signature of enhanced Andreev reflection at the interface of both conventional (for example, Nb) and high-temperature (for example, YBCO) superconductor–semiconductor (for example, n-InGaAs, and n-GaN) junctions is also an indication of the direct injection of Cooper pairs into SLEDs^{111–113}. In the above-discussed experimental reports, the enhanced emission below T_c in superconductor-coupled p–n junctions and QDs was attributed to modifications of the density of states of the host material by the parent superconductor and Cooper-pair recombination. A recent study showed two-photon emission in a Nb/GaAs/AlGaAs SLED¹⁵. The structure of the device and the measured temperature-dependent conductivity are shown in Fig. 4c, with Cooper-pair injection observed

as an enhancement in the electrical conductance below T_c . To effectively demonstrate photon correlations, uncorrelated one-photon emission should be minimized. In SLEDs, high currents significantly increase one-photon emission due to the low injection ratio of Cooper pairs to single electrons, necessitating low-current operation. Given that the voltage drop across the p–n junction is significantly larger than that across the superconductor–semiconductor junction, the electron quasi-Fermi energy gradient within the superlattice remains small. This allows the application of higher voltages while keeping the quasi-Fermi level within the superlattice miniband, ensuring high Cooper-pair injection efficiency via resonant tunnelling and minimizing single-electron injection. The injection ratio of Cooper pairs to single electrons remains high even at increased currents, enabling stronger emission. However, as higher currents lead to greater Joule heating, pulsed operation is needed, with pulse widths of $\sim 100\ \mu\text{s}$ to prevent the distortion of the photon-pair correlation $g^{(2)}(\tau)$. The statistical properties of the two-photon emission can be characterized through correlation measurements $g^{(2)}(\tau, T) = \langle a_1^\dagger(t)a_1^\dagger(t+\tau)a_2(t+\tau)a_2(t) \rangle / \langle a_1^\dagger(t)a_1(t) \rangle \langle a_2^\dagger(t+\tau)a_2(t+\tau) \rangle$ using the Hanbury Brown–Twiss set-up. Here τ is the time interval between photon detections, and $a_1^\dagger(t)$ and $a_1(t)$ represent photon creation and annihilation operators, with $\langle \rangle$ denoting a time and ensemble average. Below T_c , a peak in $g^{(2)}(\tau=0)$ with a width of around 0.5 nS that varies with temperature is attributed to photon-pair correlations (Fig. 4c, right); spectral broadening and shifts below T_c , along with enhanced emission at longer wavelengths for bias currents and temperatures below I_c and T_c , are also indicative of two-photon emissions¹⁵.

Future prospects

This Review examined on-chip, electrically driven coherent superconducting light sources as critical quantum hardware, especially for cryogenic applications (Fig. 5). I showed the promise of superconducting photon sources operating at microwave, millimetre-wave, THz and optical frequencies for applications in quantum information processing. Quantum technologies (including communication, computing, sensing, navigation, metrology and future space and environmental monitoring technologies) demand the development of chip-integrated electrically driven coherent, low-temperature-compatible, continuous-wave and pulsed photon generators with robust stability, high signal resolution, high repetition rate, exceptional spectral purity, excellent efficiency and extremely low heat load. The sources should be able to provide fine control over the frequency, intensity and phase, while also being compatible with cryogenic quantum circuits for monolithic integration^{114,115}. In quantum computers, replacing larger-scale d.c. or r.f. cabling and joints with densely integrated low-loss superconducting circuit modules may significantly decrease the footprint and heat load of these systems. The monolithic integration of superconducting microwave sources with quantum processors provides systematic benefits such as reduced communication latency, increased reliability and improved signal quality, allowing high-fidelity qubit state control and readout at millikelvin temperatures¹¹⁴. For quantum computing, such features are critical to facilitate scalable, error-corrected quantum computing and develop noisy intermediate-scale quantum applications. Challenges that remain to be addressed include maintaining large-scale low-power superconducting electronics at cryogenic temperatures for coherent microwave devices, which complicates system design and increases operational costs. Integrating superconducting materials with conventional electronic components also requires precise fabrication, integration and packaging techniques to ensure compatibility and optimal performance in terms of the coherence time, frequency stability and power efficiency. For THz quantum hardware, the primary challenge is the THz gap (and the lack of efficient chip-scale sources, modulators and detectors in the 0.1 to 10 THz range), which limits practical applications. Furthermore, THz waves are absorbed by atmospheric humidity, posing complications to long-distance

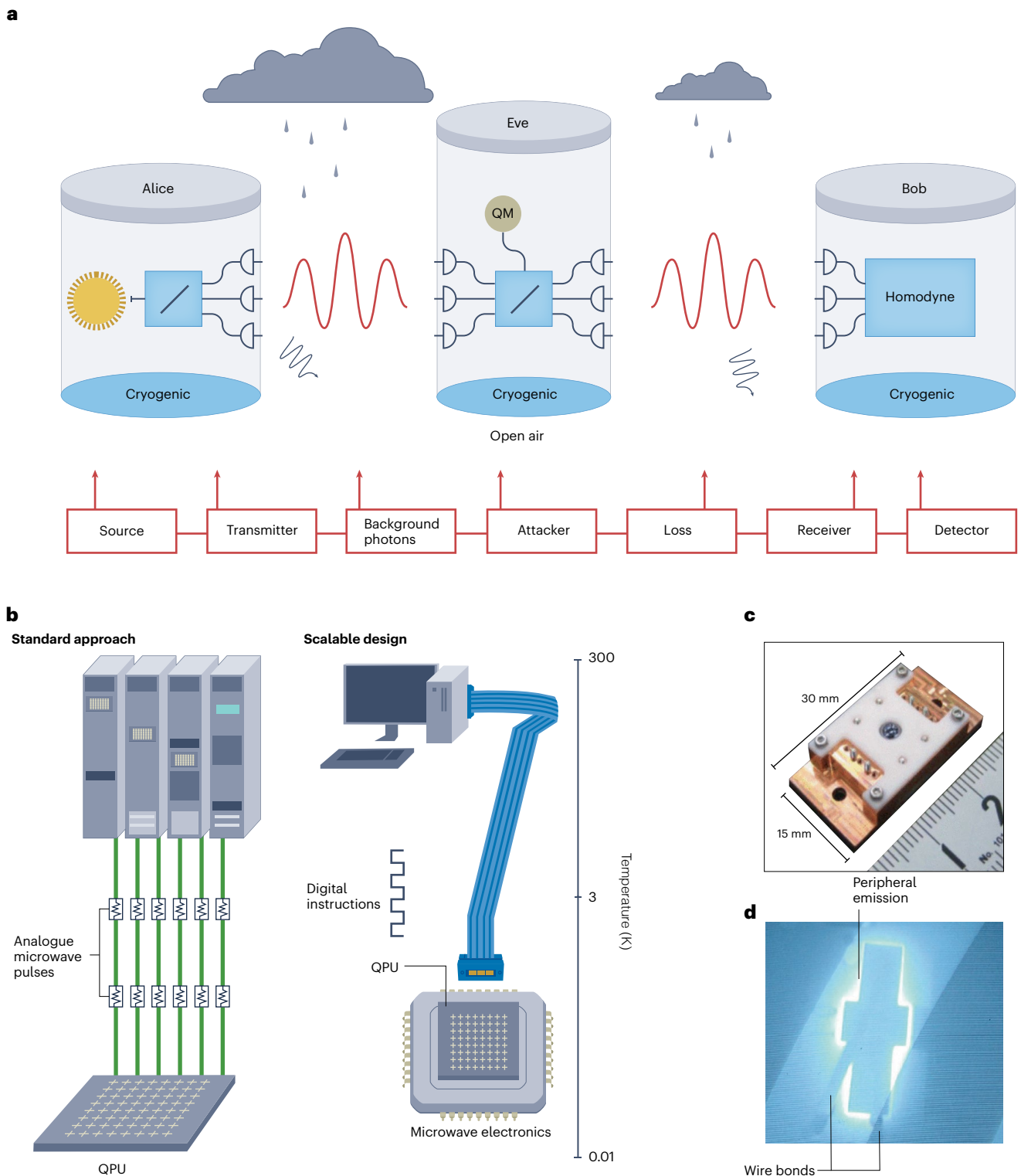


Fig. 5 | Superconducting platforms for future quantum networks. **a**, System-level architecture of a quantum communication network using cryogenic components. Alice generates thermal states, Bob performs quadrature measurements via homodyne detection and Eve stores intercepted modes in a quantum memory (QM). Signal losses include atmospheric absorption and free-space propagation effects⁵. **b**, Conceptual monolithic superconducting quantum processor that integrates a quantum processing unit (QPU), chip-scale cryogenic control electronics and microwave generators (right) compared with a standard bulky set-up (left). This architecture supports scalable and compact

quantum computing¹¹⁴. **c**, Photograph of a compact, portable superconducting THz laser module. The coherent THz source is based on IJJs, demonstrating on-chip integration¹²⁹. **d**, Optical image of a SLED based on a Nb/GaAs/AlGaAs superlattice. The device emits photon pairs in the 820–860 nm range, supporting optical-domain applications¹⁵. Figure reproduced with permission from: **a**, ref. 5 under a Creative Commons license CC BY 4.0; **b**, ref. 114 under a Creative Commons license CC BY 4.0; **c**, ref. 129, IOP; **d**, ref. 15 under a Creative Commons license CC BY 4.0.

communication. In the short term, the focus will be on advancing compact, tunable, electrically driven, electrically tunable, chip-scale coherent THz sources, modulators and detectors to bridge this gap. The integration of superconducting THz devices and circuits into existing communication and imaging systems will further validate their practicality. Long-term goals include adopting current constraints to facilitate the widespread implementation of THz technologies across various fields, such as communications, computing, sensing, medical imaging and security, with rich advantages over existing solutions. In quantum communication, for example, producing and sending high-fidelity, distributed multi-qubit entanglements deterministically will facilitate large-scale quantum communication and computational networks with high state-transfer fidelities¹¹⁶. Robust quantum microwave, THz and optical modules capable of coherently linking many quantum processors on a chip, inside a cryostat or across multiple interconnected cryostats are yet to be developed¹¹⁷. Advances in electrically driven full-spectrum quantum integrated devices show potential for developing future quantum networks and enabling long-distance quantum communication with high-fidelity, low-loss communication links^{118–120}. Previous experiments with microwave quantum communication have been carried out over distances ranging from 0.5 to 30 m (refs. 121,122). Recent efforts using superconducting parametric devices have enabled generation and single-shot quadrature detection for continuous-variable microwave quantum key distribution, demonstrating the feasibility of secure microwave quantum communication in both open-air (up to ~80 m) and cryogenic (exceeding 1,000 m) environments¹²³. In comparison, the development of superconducting coherent sources operating at THz frequencies with efficient cryogenic compatibility may allow longer-range ultrafast wireless communications networks^{5,80}. For optical applications, electricity-driven compact SLEDs with hybrid nanostructures that have improved performance have arisen as an important solution¹⁵. For SLEDs, challenges include achieving high-quality contacts between superconducting materials and semiconductor structures through interface engineering approaches, as differences in lattice structures and thermal properties complicate this process (especially for wafer-scale production and large-scale integration), and packaging. The other immediate goals include improving the emission efficiency and coherence properties, which would make SLEDs more practical for quantum applications such as on-demand photon generation. In the long term, integrating SLEDs into quantum communication and computing systems will be vital to create high-fidelity photon sources. Advances in these technologies are essential for developing quantum information processing and realizing a scalable quantum internet. Improved nanofabrication techniques have made it possible to manufacture devices and circuits that benefit from the properties of both semiconductors and superconductors that incorporate complex structures at the nanoscale^{124–132}. However, significant developments in the monolithic integration of semiconductor and superconductor materials, interfacial engineering and packaging are required to reach industry-level production appropriate to implement and deploy them in practical quantum systems and networks for robust quantum information processing.

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Competing interests

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